2. Big Bang to Now

1 Science fiction loves to ignore this. There's an early episode of *Star Trek: The Next Generation* in which they accidentally travel a billion light-years in a few seconds and the place they end up in is some kind of abyss of shimmering blue energy and thought, which, if it really existed, we'd totally be able to see in telescopes.

2 We can thank relativity for this. Special relativity says time passes more slowly for us when we are moving quickly; general relativity says it slows down when we are close to a massive object.

3 When Doc Brown in *Back to the Future* proclaimed, "You're not thinking fourth-dimensionally!," he was talking to you.

4 I'm being a bit flippant here but this is a rather important point. So far, in physics, most of what we've done is describe the universe with mathematical constructions we call *models*, and use experiments and observations to test and refine those models, until we arrive at a model that seems to fit the observations better than any of the others. And then we start trying to break the model. It's not that we just trust that math is fundamental to the universe, it's that there doesn't seem to be any other way to approach these things that makes any sense.

5 "Our WHOLE universe was in a hot dense state, then nearly 14 billion years ago expansion started …" Yes, the Barenaked Ladies got it right: the beginning of the theme song for the TV show *The Big Bang Theory* is actually a very good summary of the theory itself.

6 Of course, this is before "years" were a thing, because it is before there was a planet orbiting a star, defining a unit of time. But we can take our own units and extrapolate back and just label all the seconds as they count up to years and give them numbers, for our own convenience.

7 I just now invented this term and I'm feeling very proud of myself.